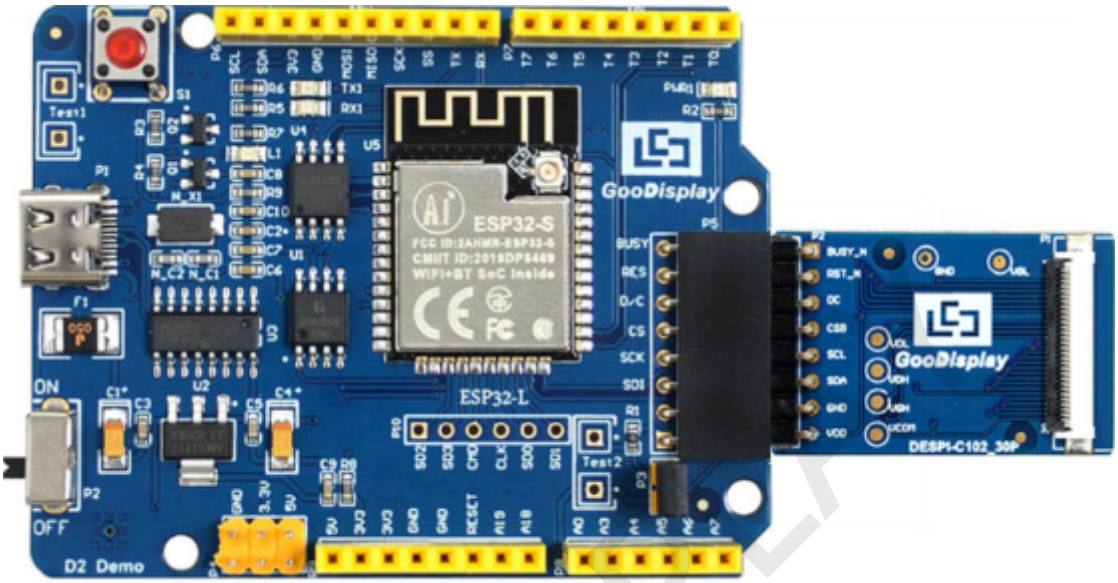


EPAPER DISPLAY EVALUATION KIT FOR ESP32, ESP32-L(C102)

GooDisplay®

Dalian Good Display Co., Ltd.

Product Specifications



Customer	**Standard**
Description	**Evaluation Kit For E-paper Display**
Model Name	**ESP32-L(C102)**
Date	**2023/10/17**
Revision	**2.0**

	Design Engineering		
	Approval	**Check**	**Design**
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1.Overview

The ESP32-L (C102) Development Kit supports programming using the Arduino development platform. This development kit helps developers accelerate and streamline the development of e-paper display projects. It is designed for SPI interface e-paper displays and can implement the refresh functions for the GDEW0102I3F and GDEW0102T4 e-paper models.

The ESP32-L (C102) Development Kit consists of two parts: the ESP32-L mainboard and the DESPI-C102 adapter board.

The ESP32-L (C102) Development Kit is intended solely for driving e-paper displays. Other functionalities such as WiFi require users to develop independently according to their project requirements.

2. Structure Specification

Parameter	Specification
Model	ESP32-L（C102）
Platform	Arduino
Dimension	MotherBoard：70mmx54mm(ESP32-L) Adapter：33.8mmx22.2mm(DESPI-C102)
Power Interface	Type-C
Example Code	Available
Operating Temp.	-20℃~70℃(-4℉~158℉)
Main Function	Learn to drive ePaper display; Test and evaluate ePaper display; Support secondary development
Additional Function	Type-C,LED indicator, Reset button, Font chip, Flash chip, Current detection

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3. Diagram

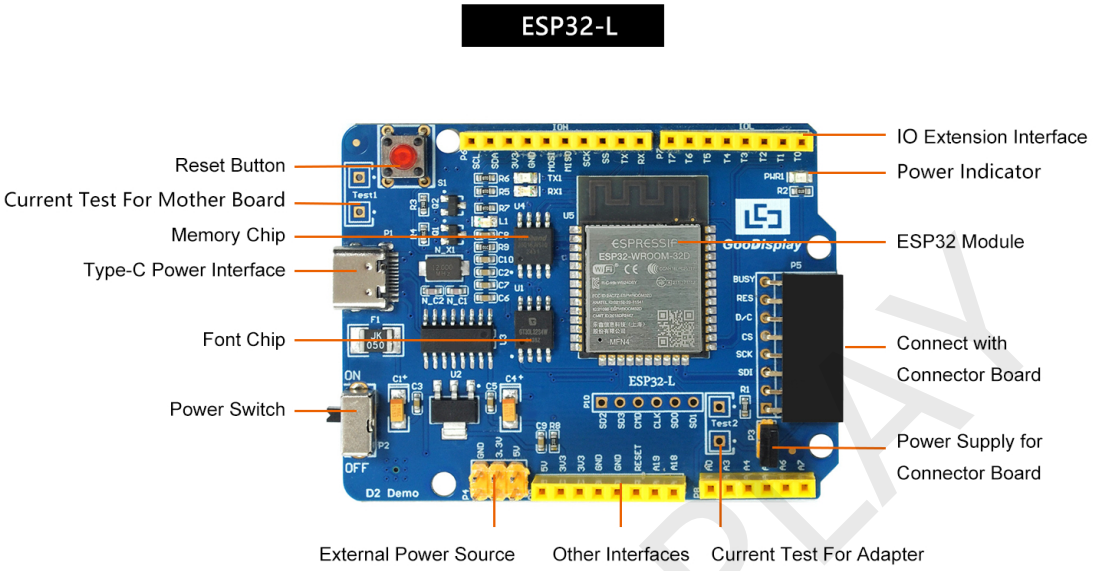
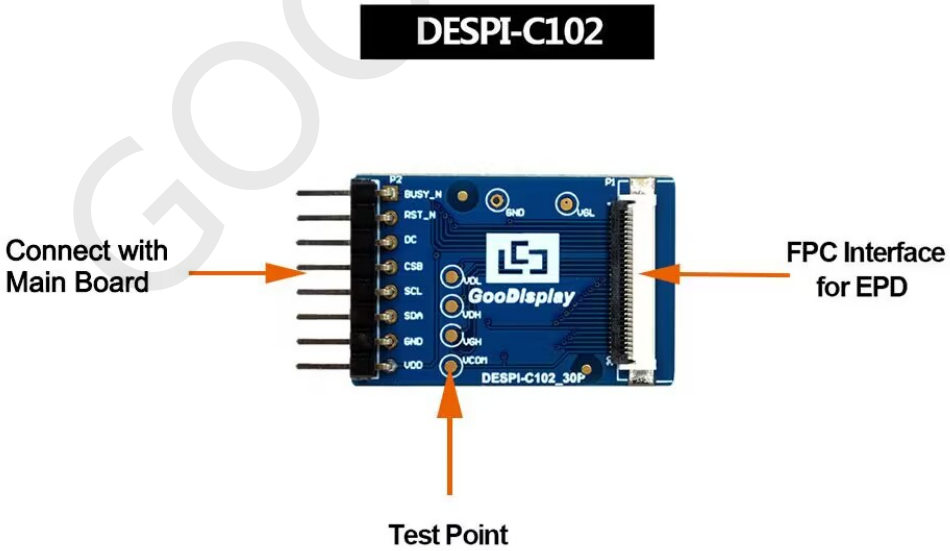


Figure 1 : ESP32-L



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3.1 Power Supply

The input voltage is DC5V, which is powered by the Type-C interface.

3.2 USB to serial port transmission

This development board uses USB to serial port communication. Users should install CH340 driver on computer before downloading program.

3.3 P3 short-circuit jumper

The P3 jumper cap controls the power supply to the adapter board, specifically the e-paper power. It must be connected during use.

3.4 Current measurement

This development board supports current measurement for both the mainboard and the adapter board.

1. Mainboard current measurement: Set the power switch to OFF and connect the ammeter in series to TEST1.
2. Adapter board current measurement: Set the power switch to ON, remove the P3 jumper cap, and connect the ammeter in series to TEST2. After testing, reconnect the P3 jumper cap.

3.5 I/O port extension

This development board led out the digital D0~D12 and the analog AO for development.

3.6 LED indicator light

There is a indicator light reserved for developing.

3.7 Reset key

This development board contains a reset key for users operation.

3.8 Expanded Functions

Built-in Chinese font chip GT30L32S4W.
Built-in data storage chip W25Q16.

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4. Connection Mode and RESE Selection

4.1 Connection between ePaper display and development board
Connect the mainboard and adapter board as shown in Figure 3. Connect the e-paper FPC to the adapter board ’ s connector as shown in Figure 4 (pay attention to the orientation of the e-paper connection).

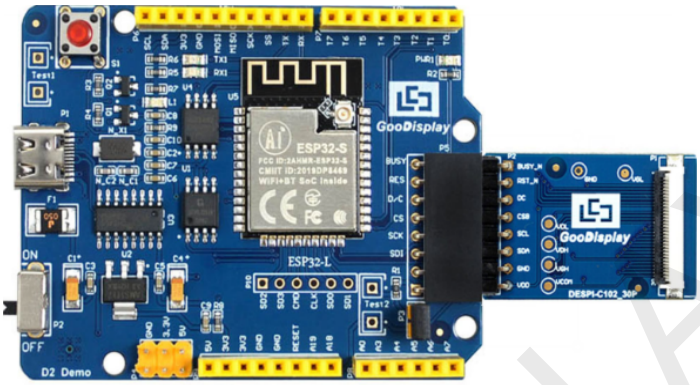


Figure3 Mainboard and Adapter Board Connection Diagram

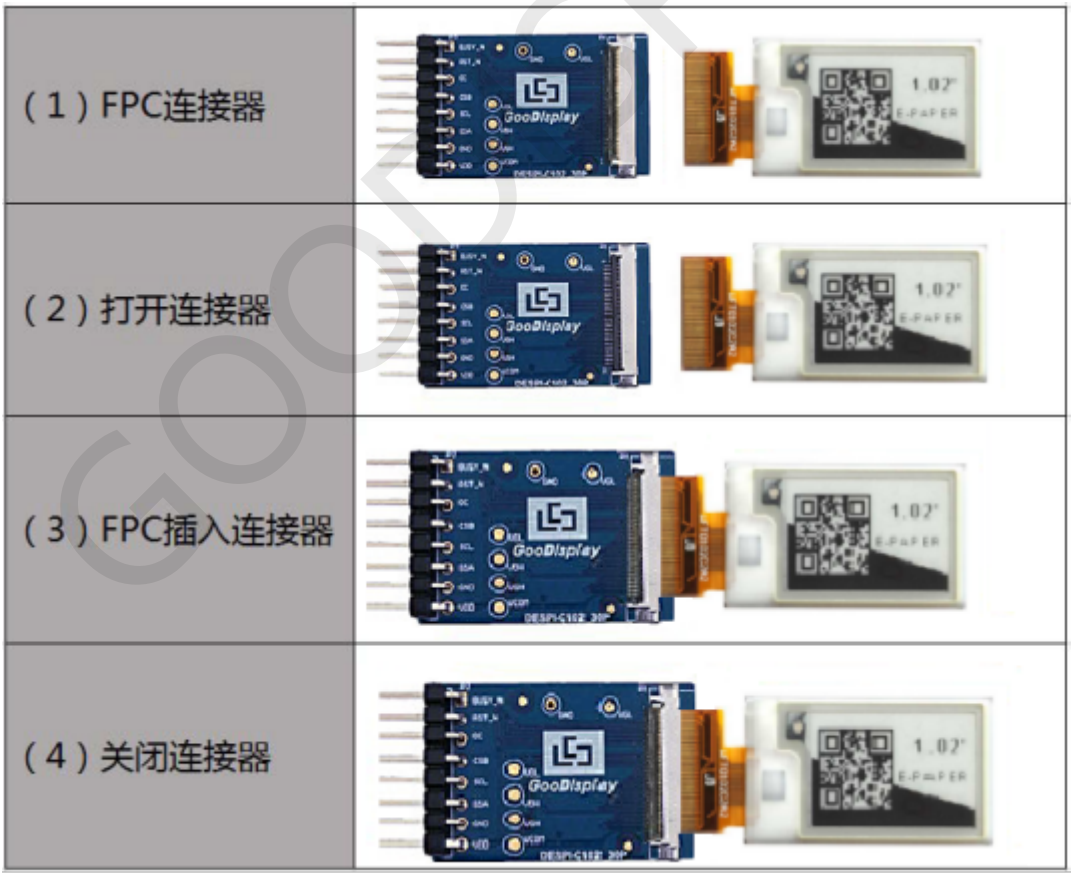


Figure 4 E-paper and Adapter Board Connection Diagram

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5. Program Downloading

This development board uses a serial port for program downloading and requi res the Arduino IDE, a Type-C data cable, and the CH340 driver. The operatio n steps are as follows:

5.1 Installing Arduino IDE and ESP32 Development Board

1) Download the Arduino IDE software from the official Arduino website: [ww w.arduino.cc](www.arduino.cc). (Using version V2.3.3 as an example.)
2) In Arduino IDE, go to Tools > Board > Boards Manager.
3) In Boards Manager, search for ESP32, select the appropriate version, and c lick INSTALL.

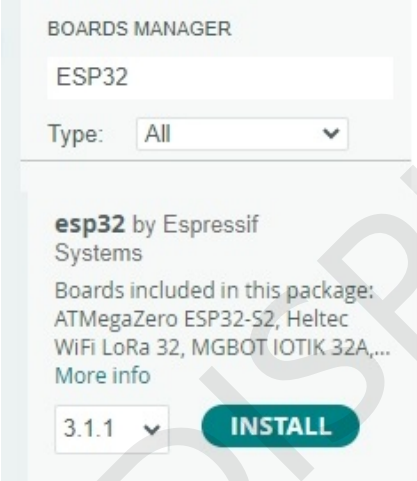


Figure 5: ESP32 Development Board Library Installation Diagram

4) After the ESP32 development board installation is complete, you will see the r elevant board information in Tools > Board.

5.2 Downloading the ESP32 Development Board Driver

1) Connect the USB interface of the development board to the computer using a Type-C data cable.
2) Open the ino project file in the e-paper driver program folder using Arduino I DE, as shown in the figure below.

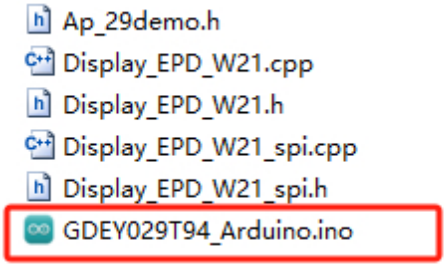


Figure 6: Opening the ino Project File

3) Steps to Download the E-paper Driver Program for the ESP32 Development Board

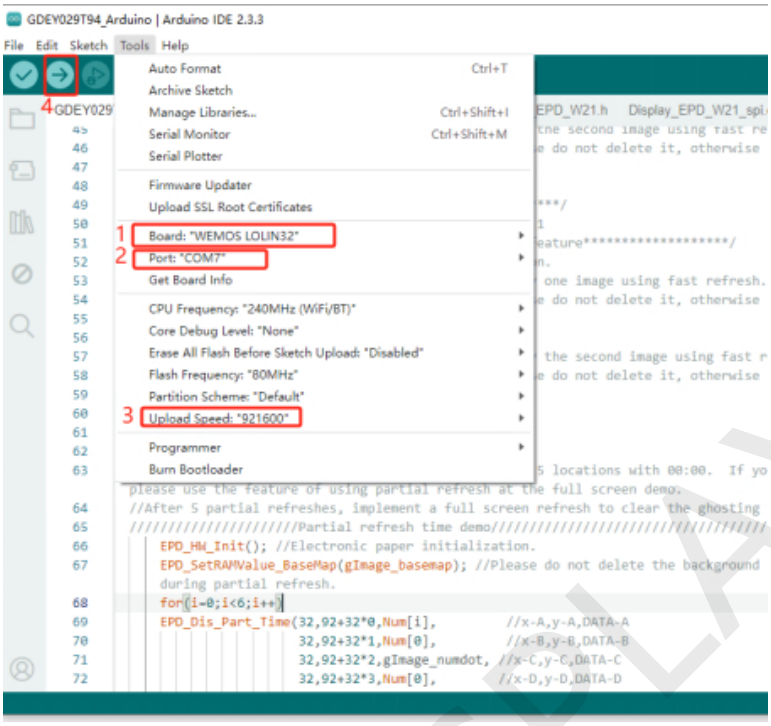


Figure 7: Program Download Operation Steps

1--Select the development board model "WEMOS LOLIN32".
2--Select the COM port.
3--Select the serial baud rate "921600".
4--Download the program to the development board.

4) After a successful download, first power off the development board, connect the e-paper display to the adapter board, then power on again. The e-paper will display th e image normally.

5.3 Using the ESP32epdx E-paper Official Library

Users can search for "ESP32epdx" inManage Librariesin Arduino IDE and install this official e-paper library. This library includes common Good Display e-paper drivers, s upports GUI functionality, and will continuously add new e-paper application feature s, such as controlling the e-paper display via Bluetooth, WiFi, etc.

Github: https://github.com/gooddisplayshare/ESP32epdx

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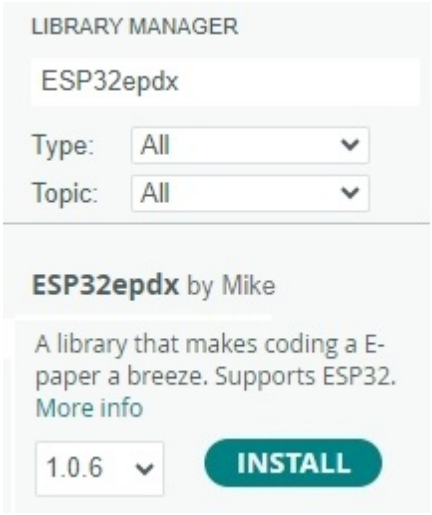


Figure 8: ESP32epdx Official Library Installation Diagram

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